

HOTEL BOOKING DEMAND

Executive Dashboard Portfolio

Dynamic Pricing & Booking Optimization · Analytics Team Lead · July 2026

Repo: github.com/KANGuyen2802/Hotel-Booking-Demand

Stack (dashboard): Python, pandas, Streamlit, Plotly, HTML/JS, Power BI (in progress)

1. Executive Summary

This portfolio piece documents the executive dashboard layer of the Hotel Booking Demand project: decision-support views that translate 82,811 City and Resort bookings into RevPAR, cancellation, and pricing what-if intelligence for GM / Revenue / Finance stakeholders.

The dashboards sit on a star-schema CSV layer and are delivered in three channels: Streamlit local web (delivered), HTML/JS local web (delivered), and Power BI executive pack (in progress).

Analytics behind the screens produced an asymmetric pricing playbook (City Peak can harden ADR; Resort Peak must not take pure +ADR shock), a cancellation-tier booking workflow, and modeled annualized uplift scenarios of roughly €10k / €59k / €70–85k on a ~€2.84M portfolio proxy—reported as floor / base / upside, not live P&L.

2. Project Scope & Objectives

Scope

- Build interactive executive dashboards for two properties (City Hotel, Resort Hotel) over Jul 2015–Aug 2017.
- Expose portfolio KPIs (bookings, revenue, ADR, occupancy, RevPAR, cancel rate) with hotel/year filters.
- Support deeper exploration: seasonality, segment/country mix, cancellation drivers, and pricing what-if simulation.
- Align visual narrative with the recommend-only pricing playbook and pilot governance (shadow → pilot → scale).

Objectives

1. Decision clarity: Make City vs Resort demand asymmetry visible in one executive surface.
2. Operational readiness: Give RM / FO / Finance a shared KPI language for the 16-week pilot.
3. Metric integrity: Compute RevPAR / ADR / occupancy with an explicit, auditable formula.
4. Multi-channel delivery: Local web for analysts + static HTML for lightweight sharing + Power BI for stakeholder publish (in progress).

Out of scope

- Live OTA price push / RMS automation.
- Causal A/B in production (dashboards support shadow/pilot monitoring design only).

3. Business Questions Addressed

#	Business question	Dashboard response
Q1	How do City and Resort differ on RevPAR, ADR, occupancy?	Overview + RevPAR with hotel/year filters
Q2	Where does cancellation risk concentrate?	Cancellation Analysis drivers
Q3	What if we move ADR in a controlled band?	Pricing Simulator what-if
Q4	Which markets/segments drive revenue vs cancel?	Segment & country mix
Q5	Ready to pilot asymmetric BAR rules?	KPI pack: Δ RevPAR, Δ cancel, walk, Direct

4. Design Thinking Approach

1. Empathize — Stakeholders need one story: raise revenue without uncontrolled cancel/walk/OTA conflict.
2. Define — Problem framed as property-asymmetric demand, not a single portfolio elasticity.

3. Ideate — Separate pages for Overview / RevPAR / Cancellation / Pricing; dual local-web channels + Power BI.
4. Prototype — Streamlit multipage app + HTML static dashboard on the same star-schema facts.
5. Test / refine — Filters, design tokens, recommend-only framing (no auto-publish of extreme +21% raises).

Design principle: show the trade-off (ADR vs volume vs cancel risk), not a single “optimal price” number.

5. Dataset Overview

Attribute	Detail
Source file	hotel_bookings_v5.csv (cleaned panel)
Grain	1 row = 1 booking
Volume	82,811 bookings
Properties	City Hotel, Resort Hotel
Time window	Arrival months 2015-07 → 2017-08
Overall cancel rate	28.12%
High-risk cancel tier	$P \geq 0.55 \rightarrow$ realized cancel ~64% (n = 28,560)

6. Data Model Design

Schema Choice

Star schema (CSV exports) chosen over a single denormalized flat file for dashboards.

Layer	Files	Grain
Fact (monthly)	revpar_monthly.csv	Hotel × calendar month
Fact (booking)	hotel_bookings_normalized.csv	1 booking
Dimensions	dim_hotel, dim_date, dim_country, ...	Lookups for Power BI / joins

Why star schema: clear separation of monthly RevPAR KPIs vs booking-level cancel/segment analysis; stable grain for Power BI; rebuildable via scripts/build_star_schema_v5.py (pandas).

7. Dashboard Architecture

hotel_bookings_v5.csv → build_star_schema_v5.py → star-schema CSVs → Streamlit (dashboard/) | HTML JSON export (dashboard-html/) | Power BI (dashboard-powerbi/, in progress).

Streamlit pages

Page	Purpose
Overview	CEO KPIs, revenue/bookings, status, segment, country
RevPAR	ADR × occupancy, seasonality, monthly panel
Cancellation Analysis	Deposit / channel / segment / lead-time drivers
Pricing Simulator	Monthly what-if ADR moves (recommend-only)

Shared UX: sidebar Hotel/Year multiselect; weighted portfolio KPIs; design tokens from design-system/hotel-booking-demand/.

8. Key Findings & Strategic Recommendations

Key findings

1. City Peak tolerates controlled ADR increase: +10% ADR → +2.3% RevPAR; Peak back-test win-rate 100% in the evaluated window.
2. Resort Peak does not: same +10% ADR → -2.1% RevPAR — pure ADR shock is policy-forbidden.
3. Pure analytic optima (~+21% City RAISE) are too aggressive; operable ±15% bands plus dual-objective softens BAR by ~7–8%.
4. Cancellation is manageable by tier: High-risk bookings feed buffer → Direct @ BAR instead of OTA dump.

5. Narrow rules back-tested 2015–2016 with go=True ($\Delta\text{RevPAR} \geq 0$; $\Delta\text{cancel} \leq +1$ pp) for both properties.

Strategic recommendations

ID	Action
R1	Asymmetric playbook: harden City Peak; HOLD Resort Peak; Resort Low CUT $\sim -5\%$
R2	Never publish extreme +21% as live BAR; keep ensemble band + risk controls
R3	Route bookings by cancel tier (Low / Med / High $\rightarrow$ Direct)
R4	Prefer Direct refill at recommend BAR before OTA dump
R5	16-week staged pilot with kill switches (walk $>5\%$ /week; cancel $>+1$ pp)

Modeled impact (proxy): Conservative $\sim \text{€}10\text{k}$ · Full in-band $\sim \text{€}59\text{k}$ · Direct/buffer upside $\sim \text{€}70\text{--}85\text{k}$ on $\sim \text{€}2.84\text{M}$ base.

9. Technical Implementation Notes

Topic	Implementation
Runtime data access	dashboard/lib/db.py — pandas reads star-schema CSVs (cached)
Charts	Plotly (Streamlit); Chart.js / custom JS (HTML)
HTML data refresh	dashboard-html/_export_data.py $\rightarrow$ JSON aggregates
Local HTML serve	python -m http.server (no file:// fetch)
Power BI	dashboard-powerbi/ — PBIX in progress
Dependencies	streamlit, plotly, pandas
Rebuild facts	python scripts/build_star_schema_v5.py

10. Metric Calculation Integrity Notes

Metric	Formula / rule
Occupancy (hotel-month)	$\text{mean}(1 - \text{is_canceled})$
ADR (hotel-month)	$\text{mean}(\text{adr})$ where $\text{is_canceled} = 0$
RevPAR (hotel-month)	$\text{ADR} \times \text{Occupancy_Rate}$
Cancel rate (filter)	$\text{canceled_bookings} / \text{total_bookings}$
Overview ADR/Occ/RevPAR	Booking-weighted across months in active filter

Integrity controls

- Monthly RevPAR is recomputed from booking facts, not an opaque external field.
- City and Resort are never blended into one elasticity on decision pages.
- Pricing Simulator is what-if / recommend-only; it does not write prices to channels.
- ROI figures in reports are counterfactual proxies—not committed P&L.

Known limitations

- Panel ends 2017-08 — recalibrate before live scale.
- No production RCT yet — shadow ≥ 2 weeks required before price pilot.
- Power BI executive pack is in progress; Streamlit and HTML are current interactive deliverables.

Portfolio document for the dashboard workstream of Hotel Booking Demand · July 2026